

Data Scientist Task Guide

Geographic Disparities in ACR-Accredited Cardiac Imaging Capacity

Prepared for: Geospatial Analyst / Data Scientist

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1. What Is This Project About?

In simple terms: **we want to find out if quality heart-scan facilities are fairly distributed across the US, or if they are clustered in rich urban areas while poorer rural communities have none.**

Key Terms Explained:

- **ACR Accreditation** — A quality stamp from the American College of Radiology. Facilities with this stamp meet high standards and are recognized by insurance companies.
- **Cardiac MRI / Cardiac CT** — Two types of advanced heart-imaging scans. We study each separately.

- **Choropleth Map** — A map where areas (counties) are colored based on a value (e.g., number of facilities). Darker = more, lighter = fewer.
- **SVI (Social Vulnerability Index)** — A score (0 to 1) measuring how disadvantaged a community is. Higher = more vulnerable.

What we will produce:

1. A clean dataset linking every US county with its facility counts and socioeconomic data
2. A colored map of the US showing where heart-imaging facilities are concentrated
3. A summary table comparing access across vulnerable vs. non-vulnerable communities
4. Statistical models measuring how strongly community disadvantage relates to imaging access

2. Datasets Summary

Datasets Already Provided (in the `data/` folder)

File	What It Contains	Records	Key Info
<code>ACR_Cardiac_Imaging_Sites.xlsx</code>	ACR-accredited cardiac imaging facilities (MRI + CT), filtered and ready to use	2,273	725 Cardiac MRI sites + 1,548 Cardiac CT sites. Use the sheet named "Cardiac MRI + CT (PRIMARY)"
<code>acr_dataset.xlsx</code>	Full ACR facility extract (all modalities, not just cardiac)	37,328	Reference only — you do NOT need this for the main analysis

Datasets You Need to Download

#	Dataset	Where to Get It	What It Gives You
1	HUD ZIP–County Crosswalk (Q1 2026)	huduser.gov/portal/datasets/usps_crosswalk.html	Converts ZIP codes → county FIPS codes
2	CDC Social Vulnerability Index (2022)	atsdr.cdc.gov/placeandhealth/svi/	Vulnerability score for each county
3	USDA Rural-Urban Continuum Codes (2023)	ers.usda.gov/data-products/rural-urban-continuum-codes	Classifies counties as urban or rural
4	Census ACS Population Data (2019–2023)	census.gov/data/developers/data-sets/acs-5year.html	Population counts (total + adults aged 45+)
5	Census TIGER/Line County Shapefiles (2023)	census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html	County boundary shapes for the map

3. What You Need To Do (Step-by-Step)

Follow these steps in order. Each step builds on the previous one.

STEP 1 Set Up Your Environment

- Create a Python environment (virtual env or conda)
- Install all libraries listed in Section 4
- Create folders: `data/raw/` , `data/processed/` , `output/` , `code/`

STEP 2 Download the External Datasets

- Download all 5 datasets listed above
- Save them in `data/raw/`
- For the Census API, get a free API key at api.census.gov/data/key_signup.html
- **Check:** Before downloading SVI, verify if a 2024 version exists (if yes, ask project lead which to use)

STEP 3 Clean the ACR Facility Data

- Open `ACR_Cardiac_Imaging_Sites.xlsx`, sheet "Cardiac MRI + CT (PRIMARY)"
- Keep only rows where Status = "Accredited" or "Under Review" (all 2,273 rows qualify)
- Remove any facility whose Expiration Date is before May 20, 2026
- Ensure ZIP codes are 5-digit strings (pad leading zeros, e.g., "07652" not "7652")
- Separate into two groups: Cardiac MRI (modality = "MRAP") and Cardiac CT (modality = "CTAP")

STEP 4 Link ZIP Codes to Counties

- Use the HUD crosswalk to find which county each facility's ZIP code belongs to
- **Problem:** Some ZIPs span multiple counties. **Solution:** Pick the county with the highest `RES_RATIO` (where most people in that ZIP actually live)
- After this, every facility has a 5-digit county FIPS code
- **Verify:** Total still adds to 2,273 facilities

STEP 5 Build the County-Level Dataset

Create one row per US county (3,143 total — 50 states + DC, no territories). For each county, include:

- Count of Cardiac MRI facilities and Cardiac CT facilities
- SVI score + SVI quartile (divide all counties into 4 equal groups)
- RUCC code + a simple metro/non-metro flag (RUCC 1–3 = metro, 4–9 = non-metro)
- Total population and adults aged 45+
- Rates: facilities per 100,000 adults aged 45+ (separately for MRI and CT)
- A flag for counties with fewer than 1,000 adults 45+ (exclude these from rate calculations because rates would be unreliable with so few people)

STEP 6 Create Table 1 (Summary Statistics)

Build a table showing facility access broken down by:

- **Rows:** All counties, then by SVI quartile (Q1–Q4), then by metro vs. non-metro
- **Columns:** Number of counties, population, facility counts, median rate per 100k (with range)
- **Bottom row:** Spearman correlation (a measure of how strongly two things move together) between facility rate and SVI score, with p-value

STEP 7 Create Figure 1 (The Map)

A two-panel US county map:

- **Panel A:** Cardiac MRI rate per 100k adults 45+ | **Panel B:** Cardiac CT rate
- **Colors:** Blue scale (5 shades for counties with facilities) + light gray (zero facilities) + white (excluded counties)
- **Binning:** Quantile-based (each shade has ~same number of counties)
- **Projection:** Albers USA (standard US map look) with Alaska/Hawaii as insets
- **Output:** PDF (vector) + PNG (600 dpi backup)
- **Layout:** Stack panels vertically (A on top, B below)

STEP 8 Run Statistical Models

What: Negative binomial regression — a model for count data that answers: "For every 10-point increase in vulnerability, how much does facility access change?"

- Outcome = facility count per county (run separately for MRI and CT)
- Predictor = SVI score (per 10-percentile increment)
- Offset = $\log(\text{adult population } 45+)$ — this converts counts into rates
- Report: Incidence Rate Ratio (IRR) with 95% confidence interval

Sensitivity checks (re-run models with small changes to confirm results hold):

1. Drop the 23 "Under Review" facilities
2. Use SVI quartiles instead of continuous SVI
3. Run separately for metro and non-metro counties

STEP 9 Document & Deliver

- Write a README with dataset versions, download dates, and decisions made
- Comment all code clearly
- Include a `requirements.txt` file
- Package everything for delivery

4. Open Source Tools You Will Use

Core Data & Analysis Libraries

Library	What It Does (Simple Explanation)	How You'll Use It Here
pandas	Works with data tables (like Excel but in Python)	Load files, merge datasets, calculate rates, make summary tables
numpy	Math operations on arrays of numbers	Calculations, rounding, handling missing values
scipy	Scientific/statistical functions	Spearman correlation test
statsmodels	Statistical models (regression, etc.)	Negative binomial regression, model comparison
openpyxl	Reads/writes Excel files	Load the .xlsx data files

Geospatial & Mapping Libraries

Library	What It Does (Simple Explanation)	How You'll Use It Here
geopandas	Like pandas, but understands geographic shapes (counties, states)	Load shapefiles, merge data with map boundaries, draw choropleths
matplotlib	The main Python plotting library	Render the map, handle multi-panel layout, export PDF/PNG
mapclassify	Decides how to split data into color bins for maps	Create quantile and natural-breaks (Jenks) bins
pyproj	Handles map projections (how a round Earth is shown flat)	Apply Albers USA projection

Census Data Access

Library	What It Does (Simple Explanation)	How You'll Use It Here
census	Talks to the US Census Bureau API for you	Pull population data by county automatically
us	Has US state codes, abbreviations, FIPS codes built in	Filter to 50 states + DC, look up state info

Install Everything At Once

```
pip install pandas numpy openpyxl scipy statsmodels geopandas matplotlib  
mapclassify pyproj census us jupyter
```

5. Key Columns That Matter

These are the most important columns across all datasets. Focus on these.

From the ACR Dataset (Your Starting Point)

Column	Why It Matters
Zip Code	The geographic link — connects facilities to counties
modality	Tells you MRI vs CT: "MRAP" = MRI, "CTAP" = CT
Status	Must be "Accredited" or "Under Review" to be included
Expiration Date	Exclude if expired before May 20, 2026

From External Datasets

Dataset	Key Column	What It Does
HUD Crosswalk	ZIP , COUNTY , RES_RATIO	Links ZIP → County; RES_RATIO picks the right county for multi-county ZIPs
CDC SVI	FIPS , RPL_THEMES	Overall vulnerability score (0–1). This is your main predictor variable.
USDA RUCC	FIPS , RUCC_2023	1–3 = metro/urban, 4–9 = non-metro/rural
Census ACS	GE0ID , population fields	Your denominator (adults 45+) for calculating rates
TIGER/Line	GE0ID , geometry	The actual shapes of counties for drawing the map

6. Important Rules & Warnings

Do NOT fix the 5 states with zero MRI sites (Alaska, New Mexico, South Dakota, Utah, Wyoming). This is real — those states use a different accreditation system. It will be discussed in the paper's Limitations section.

Always store ZIP codes and FIPS codes as text strings, not numbers. Otherwise leading zeros get dropped (e.g., "07652" becomes "7652") and your data linkage will break.

Other rules:

- Only include 50 states + DC. Exclude US territories (Puerto Rico, Guam, etc.)
- The 23 "Under Review" facilities ARE included in the main analysis
- Nuclear Cardiology sheet is out of scope — ignore it
- A facility accredited for BOTH MRI and CT counts once in each — this is correct, not a duplicate

7. Final Deliverables Checklist

#	What To Deliver	Format	Due
1	Linked county-level analytic dataset	CSV	May 24
2	Figure 1 — Two-panel US choropleth map	PDF + PNG (600 dpi)	May 24
3	Table 1 — Summary by SVI quartile & rurality	CSV or Excel	May 24
4	Regression results (IRR, confidence intervals, model fit)	Text summary + pickle	May 24
5	All analysis code (well-commented)	.py or .ipynb	May 24
6	README documentation	Markdown	May 24

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